

ODS: Test-Time Adaptation in the Presence of Open-World Data Shift

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Problem: Adapt source models online when both covariate $D_t(X)$ and label $D_t(Y)$ distributions shift.

Introduction & Motivation

- DNNs assume i.i.d.; real-world shifts (weather, sensors) break this.
- Most TTA methods target covariate shift only; label shift often ignored.
- Label shift matters in practice (e.g., disease prevalence spikes).
- AODS: adapt online to changing $D_t(X)$ and $D_t(Y)$.



Contrast: covariate shift (feature distribution changes) vs label shift (class priors change). AODS handles both.

Overview

- Study TTA with simultaneous covariate and label shifts: **AODS**.
- Analyze classification error vs. distribution shifts.
- Propose **ODS**: decouples shifts via two modules: **Distribution Tracker (M_T)** and **Prediction Optimizer (M_O)**.
- Boosts multiple TTA baselines on **CIFAR10/100-C** under varied label-shift levels.
- Generic, efficient, and plug-and-play with existing TTA methods.

Problem Setup: AODS

- Multi-class classification with input $X \in \mathbb{R}^d$, labels $Y = [K]$.
- Deploy source model $f_{\{\theta_0\}}$ trained on balanced $D_0(X, Y)$.
- At timestamp t : observe unlabeled batch from $D_t(X, Y)$, update $\theta_t \rightarrow \theta_{t+1}$.
- Goal: high accuracy under jointly shifting $D_t(X)$ and $D_t(Y)$ at test time.

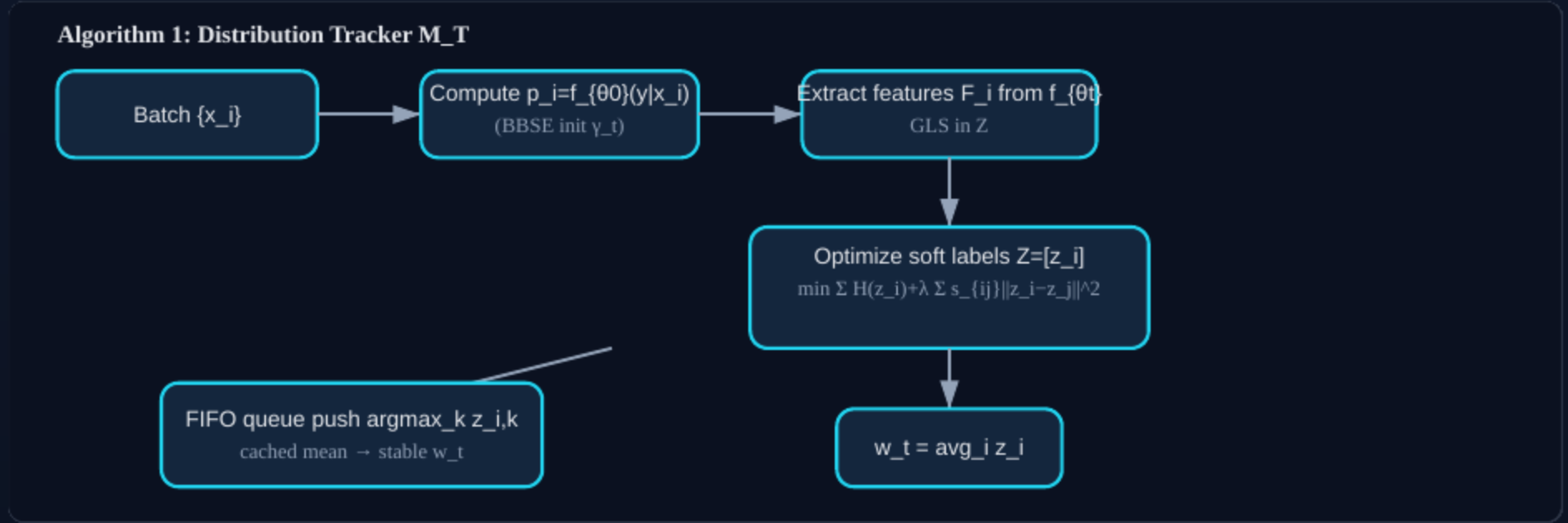
Assumption (GLS): There exists representation $Z = g^*(X)$ s.t. class-conditionals align in Z while $D_t(X)$, $D_t(Y)$ may change.

Theory: Error Decomposition (Intuition)

- Use plug-in correction to account for shifted label priors via $w_t \approx D_t(Y)$.
- Define: $BSE(\hat{Y})$ on source; $\Delta CE(\hat{Y})$ conditional error gap across domains.
- Theorem: $|\epsilon_0(\hat{Y}) - \epsilon_t(\hat{Y}^o)| \leq \text{term}(w_t \text{ vs } D_t(Y), BSE) + \Delta CE(\hat{Y}) + C$.
- Implications: better w_t estimation and robust features $\downarrow$ error gap.

Design guidance: Track w_t and adapt features; then optimize predictions to match $D_t(Y)$.

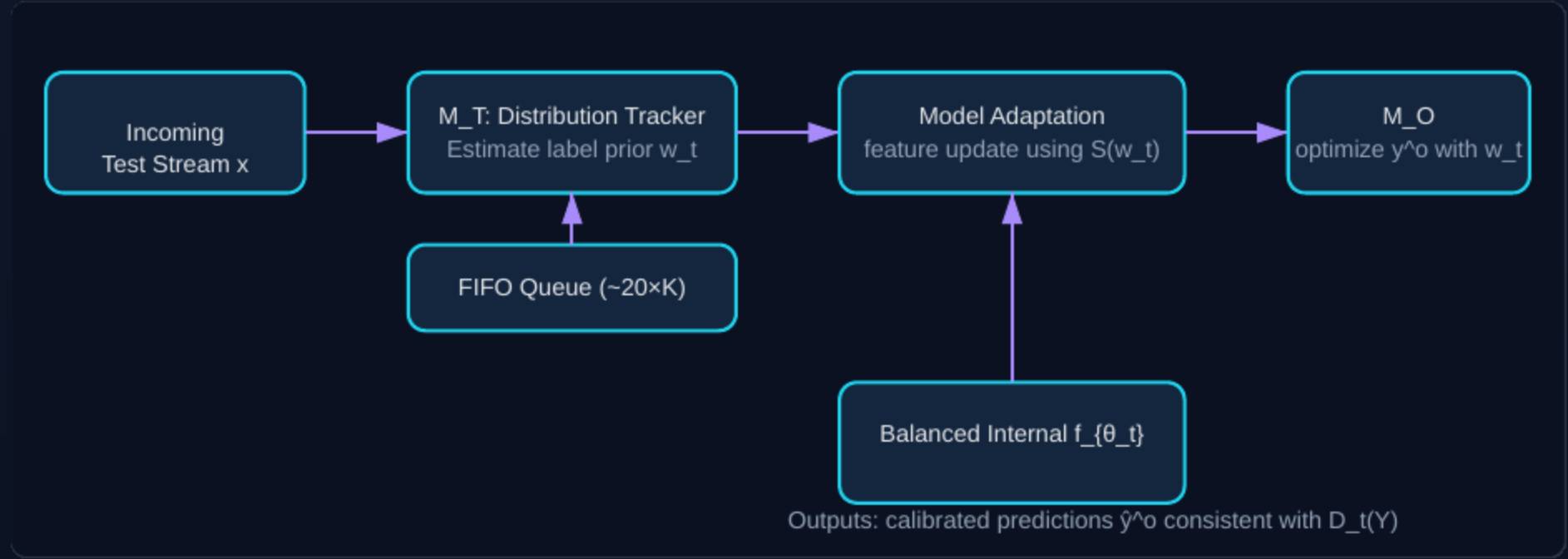
M_T: Estimating Label Distribution



- Initialize with f_{θ_0} stats; refine z_i in feature space; queue smooths w_t .

Algorithm 1. Queue update and soft-label loop yield stable w_t each timestamp.

ODS Framework



- Stream $\rightarrow$ M_T: estimate label distribution w_t (stabilized by FIFO queue).
- Adaptation: reweighted objective using $S(w_t)$ to update features.
- M_O: combine $f_{\theta_t}(Y|x)$ with w_t to output calibrated $\hat{y}^o$.

M_O: Optimizing Predictions



- Conservative ensemble with z_i for robustness to w_t noise.

Two variants: SO (statistics-based) and DO (distribution-based, robust). DO performs best empirically.

Adaptive Objective (Sketch)

Class-reweighted adaptation: $L_t = \sum_i \text{CE}(f_{\theta_t}(x_i), \hat{y}_i)$ weighted by $S(w_t) = \text{Normalize}(1 - w_t)$ per class.

Prediction calibration: $\hat{y}_o = \text{Normalize}((1 - \alpha) f_{\theta_t}(y|x) + \alpha z_i)$ or SO: $\text{Normalize}(f_{\theta_t}(y|x) \odot (w_t / D_0(Y)))$.

- Reweight by $S(w_t)$; then calibrate outputs with w_t for consistency with $D_t(Y)$.

Reweightings combats imbalance; calibration aligns predictions to current label priors.

Results: CIFAR10-C ($\gamma=10$)

METHODS	GAUSS.	SHOT	IMPUL.	DEFOC.	GLASS	MOTION	ZOOM	SNOW	FROST	FOG	BRIT.	CONTR.	ELAST.
SOURCE	14.70	18.52	15.61	56.92	31.99	68.01	63.25	82.19	72.44	76.31	92.41	23.38	72.3%
BN STATS	50.60	51.16	45.31	71.73	47.99	69.35	68.59	60.16	60.39	64.27	69.60	67.56	59.2%
TENT	53.53	60.97	59.34	63.33	47.12	65.81	68.11	55.08	55.00	58.68	63.40	49.59	46.9%
EATA	48.94	48.21	42.05	65.44	43.42	59.81	57.27	55.09	52.98	56.00	59.54	61.47	51.3%
LAME	57.99	60.15	53.07	78.83	53.04	76.67	74.90	67.81	67.30	71.94	77.05	74.84	68.5%
COTTA	57.43	60.06	56.03	66.66	52.25	66.54	66.65	58.32	58.92	60.09	64.69	55.05	59.3%
NOTE	51.90	54.57	68.38	84.29	50.53	88.97	86.21	86.15	86.68	83.27	86.48	90.64	77.8%
ODS	67.45	65.78	71.88	88.66	56.32	90.48	88.09	86.16	86.93	83.96	87.37	91.16	79.3%

ODS gains are largest on GAUSS (+15.55 over LAME) and IMPULSIVE (+3.50 over NOTE), and improves DEFOCUS/GLASS via adapting to shifted class priors (tracking w_t mitigates prior mismatch under noise/blur).

Takeaway: ODS averages 80.67% by aligning predictions with current label priors across corruptions.

Experiments: Setup

- Datasets: CIFAR10/100-C for evaluation.
- Label shift via tweak-one: $\gamma \in \{2, 5, 10\}$.
- Backbone: ResNet-18; batch size 64 at test time.
- Baselines: BN STATS, TENT, EATA, COTTA, LAME, NOTE.
- Report mean±std over 5 seeds.

Results: Robustness Across Shift Levels

CIFAR10-C (Table 2)

METHOD	$\gamma=2$	$\gamma=5$	$\gamma=10$
SOURCE	56.41±0.05	56.12±0.07	55.77±0.16
BN STATS	78.33±0.05	71.75±0.08	60.68±0.14
TENT	68.85±3.14	66.94±3.52	56.18±4.13
EATA	79.35±0.16	69.23±0.25	53.88±0.53
LAME	78.96±0.05	75.20±0.10	68.16±0.13
COTTA	81.81±0.37	73.58±0.28	60.58±0.15
NOTE	78.81±0.27	77.96±0.75	77.18±0.38
ODS	81.13±0.09	80.40±0.36	80.67±0.29

- Stable near 80–81% across γ ; competitive even vs COTTA at $\gamma=2$.

CIFAR100-C (Table 3)

METHOD	$\gamma=2$	$\gamma=5$	$\gamma=10$
SOURCE	32.71±0.15	32.71±0.18	32.75±0.14
BN STATS	52.69±0.20	52.82±0.08	52.76±0.15
TENT	40.07±2.35	51.39±0.59	52.95±0.17
EATA	43.68±18.16	45.12±15.79	48.99±7.79
LAME	52.49±0.25	52.51±0.24	52.62±0.21
COTTA	47.74±0.59	50.48±0.57	51.72±0.47
NOTE	50.34±0.11	48.41±0.33	47.06±0.35
ODS	56.86±0.18	56.43±0.21	55.83±0.23

- Consistently best across γ ; largest margins at $\gamma \in \{2, 5\}$.

Patterns: ODS shows stability on CIFAR10-C and clear superiority on CIFAR100-C, especially at stronger shifts.

Universality: Boosting Different TTA Methods

METHODS	$\gamma=2$	$\gamma=5$	$\gamma=10$
TENT	68.85 ± 3.14	66.94 ± 3.52	56.18 ± 4.13
TENT w/ ODS	69.00 ± 5.96	73.56 ± 2.85	66.03 ± 1.89
COTTA	81.81 ± 0.37	73.58 ± 0.28	60.58 ± 0.15
COTTA w/ ODS	82.11 ± 0.25	79.74 ± 0.32	74.72 ± 0.64
NOTE	78.81 ± 0.27	77.96 ± 0.75	77.18 ± 0.38
NOTE w/ ODS	81.13 ± 0.09	80.40 ± 0.36	80.67 ± 0.29

Largest average boosts: COTTA +6.16 (@ $\gamma=10$) and TENT +6.62 (@ $\gamma=10$); NOTE lags due to prior-mismatch not being explicitly corrected.

Table 4. ODS benefits all baselines; gains are strongest when base methods ignore shifting priors.

Estimating $D_t(Y)$: Accuracy

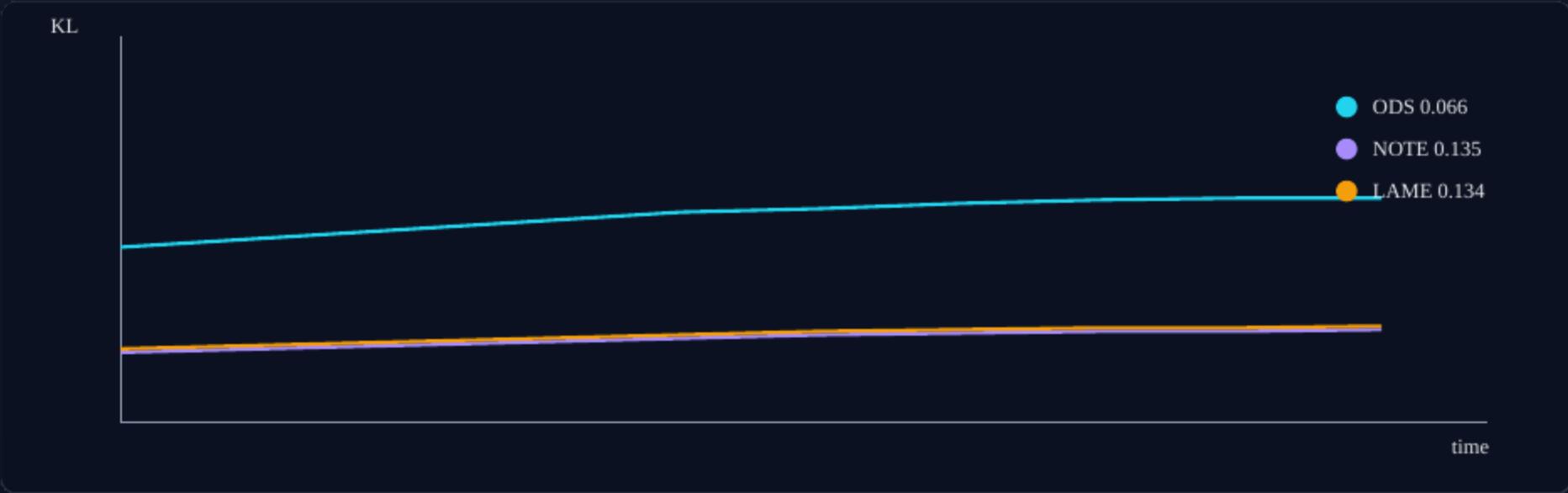


Figure 5. ODS tracks label priors most accurately over time (lowest KL to ground truth).

ODS average KL=0.066 vs NOTE 0.135 and LAME 0.134 at $\gamma=10$ → better-calibrated predictions.

Prediction Quality

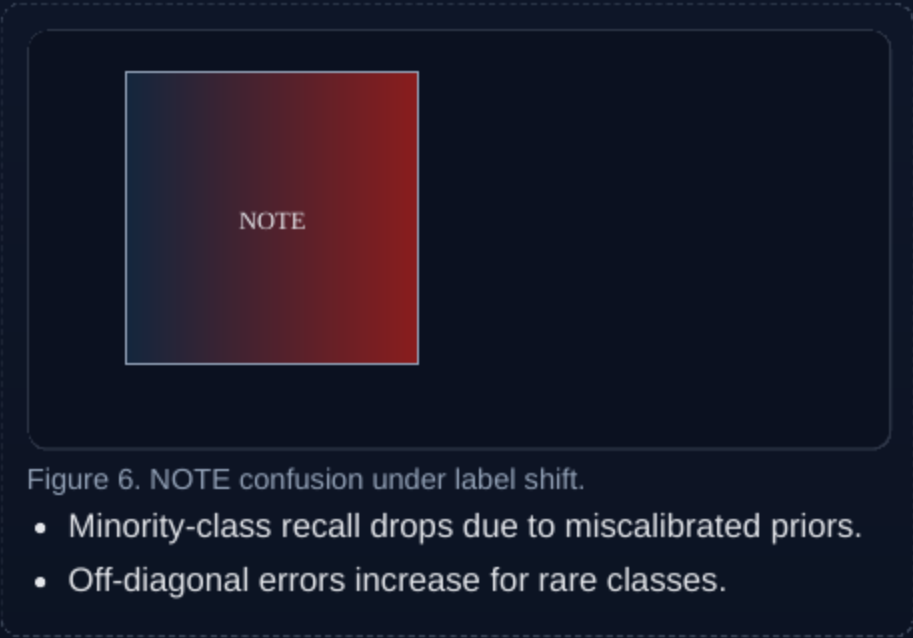


Figure 6. NOTE confusion under label shift.

- Minority-class recall drops due to miscalibrated priors.
- Off-diagonal errors increase for rare classes.

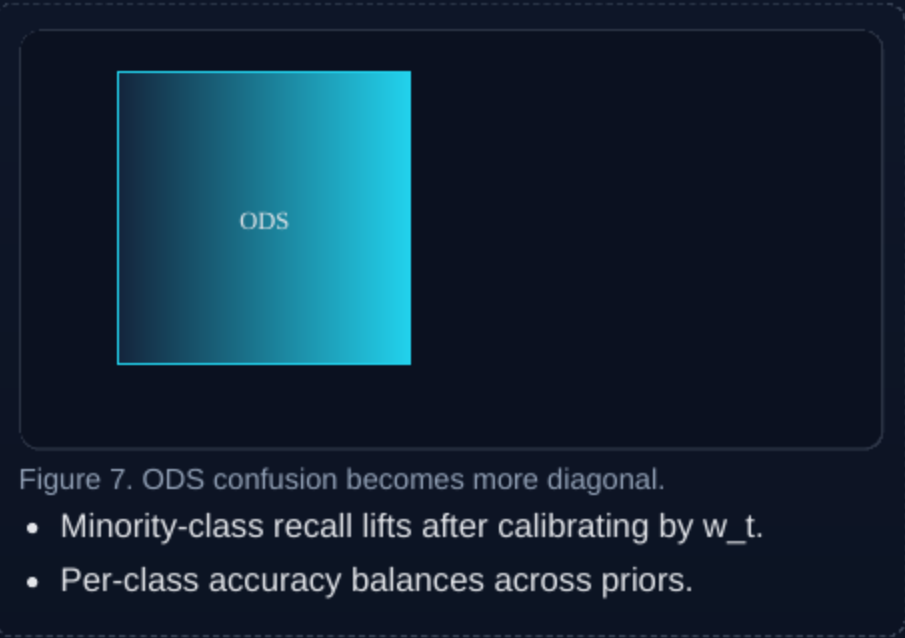


Figure 7. ODS confusion becomes more diagonal.

- Minority-class recall lifts after calibrating by w_t .
- Per-class accuracy balances across priors.

Ablation: Contributions of M_T and M_O

MODULES	TENT	COTTA	NOTE
Original	56.18 ± 4.13	60.58 ± 0.15	77.18 ± 0.38
+ M_T	58.95 ± 2.36	60.65 ± 0.31	77.20 ± 0.57
$M_T + M_O$	66.03 ± 1.89	74.72 ± 0.64	80.67 ± 0.29

Table 5. M_T alone mitigates imbalance; M_O alone (via final calibration) drives accuracy; together they reinforce each other.

Synergy: M_T supplies accurate w_t ; M_O exploits w_t to optimize outputs → largest gains.

M_O Variants and Efficiency

	NOTE	ODS w/ SO	ODS w/ DO
Performance	77.18 ± 0.38	79.50 ± 0.31	80.67 ± 0.29
Avg. Time	0.1034s (100%)	0.1150s (111%)	0.1156s (112%)

- DO adds ~12% overhead for +3.49 pts; far faster than COTTA (0.9148s).
- SO is simpler but less robust to w_t noise.

Table 6. Overhead–accuracy tradeoff favors DO in practice.

Related Work (Brief)

- TTA: TENT, EATA, BN recalibration; continual TTA: COTTA; conservative: LAME; robust non-i.i.d.: NOTE.
- Distribution shift: covariate vs label vs generalized label shift; causal/MMD/Wasserstein views.
- Online label shift: adaptation to changing priors, but often ignore covariate shift.

Comparison: ODS vs. NOTE+LAME

	γ=2	γ=5	γ=10
NOTE	78.81 ± 0.27	77.96 ± 0.75	77.18 ± 0.38
NOTE + LAME	77.32 ± 0.17	76.76 ± 0.67	78.43 ± 0.77
ODS	81.13 ± 0.09	80.40 ± 0.36	80.67 ± 0.29

- NOTE+LAME is inconsistent: helps at γ=10 only; hurts at γ=2,5.
- ODS is consistent by explicitly tracking and exploiting w_t.

Table 7. ODS' prior-tracking avoids over/under-correction seen in NOTE+LAME.

Conclusion

- ODS addresses AODS by tracking w_t and optimizing predictions while adapting features.
- Generic plug-in boosting multiple TTA methods on CIFAR10/100-C.
- Efficient with consistent gains across label-shift levels.

Limitation: Not SOTA when no label shift, but comparable. **Future:** Robust performance regardless of label shift; extend to large vision–language models.

References (Selected)

- Wang et al., 2021 (TENT); Niu et al., 2022 (EATA); Wang et al., 2022a (COTTA).
- Boudiaf et al., 2022 (LAME); Gong et al., 2022 (NOTE).
- Hendrycks & Dietterich, 2019 (CIFAR-C).
- des Combes et al., 2020; Menon et al., 2013; 2021 (label shift & plug-in).
- Lipton et al., 2018 (BBSE); Guo et al., 2020 (tweak-one shift).